



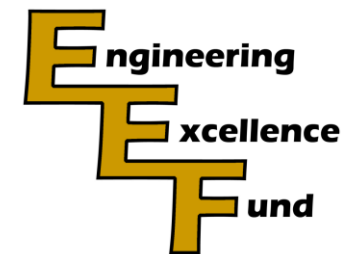
HONEYBEE SWARM FORMATION IN THE ENVIRONMENT

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WHAT IS SWARMING?

Swarming is when the queen and a group of worker bees leave the hive to establish a new hive

- Too crowded to support the current hive
- To “reproduce” and create a new hive.

Intermediate Swarm

- As the swarming group leaves, they first move to a temporary rest spot.
- The swarm forms around the queen, creating a blob of bees until they find a new home and fly there.



OUTSIDE SWARM FORMATION (EXPERIMENT)

Goal:

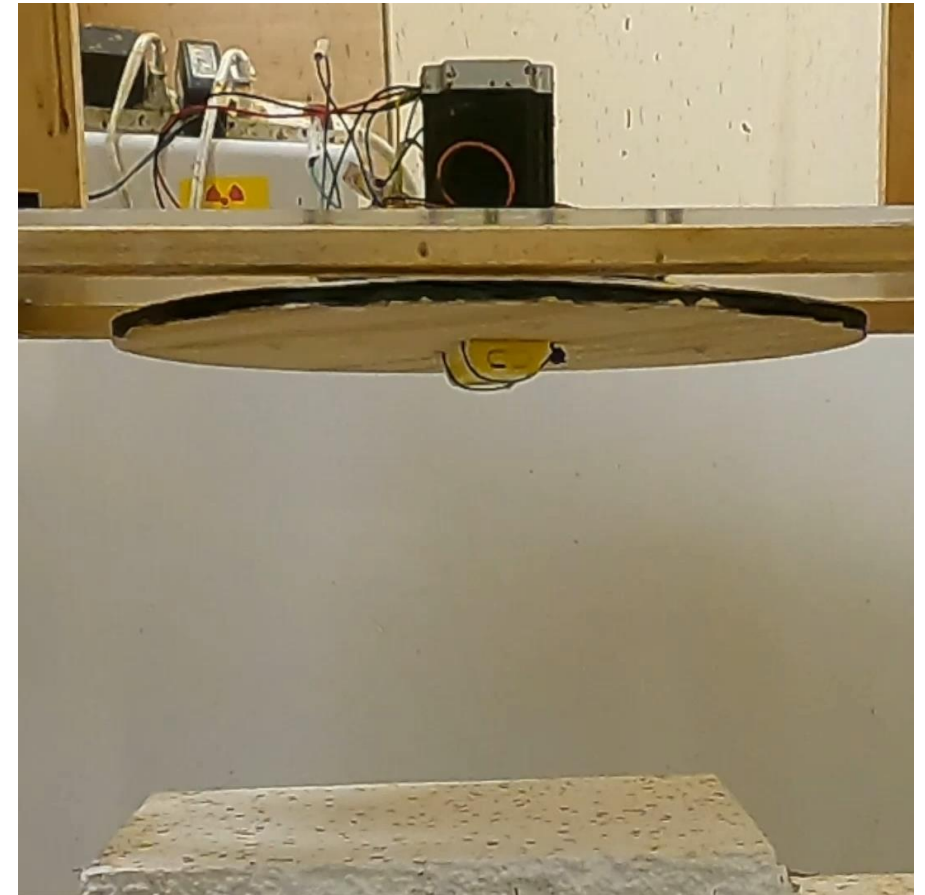
- Track how swarms form in the natural environment
 - Swarms indoors are less prone to flying and must be aided to crawl to the queen

Motivation:

- Apply in swarm robotics / intelligence

Work:

- Experiment Design & Construction
 - Data Collection
 - Image Processing
-



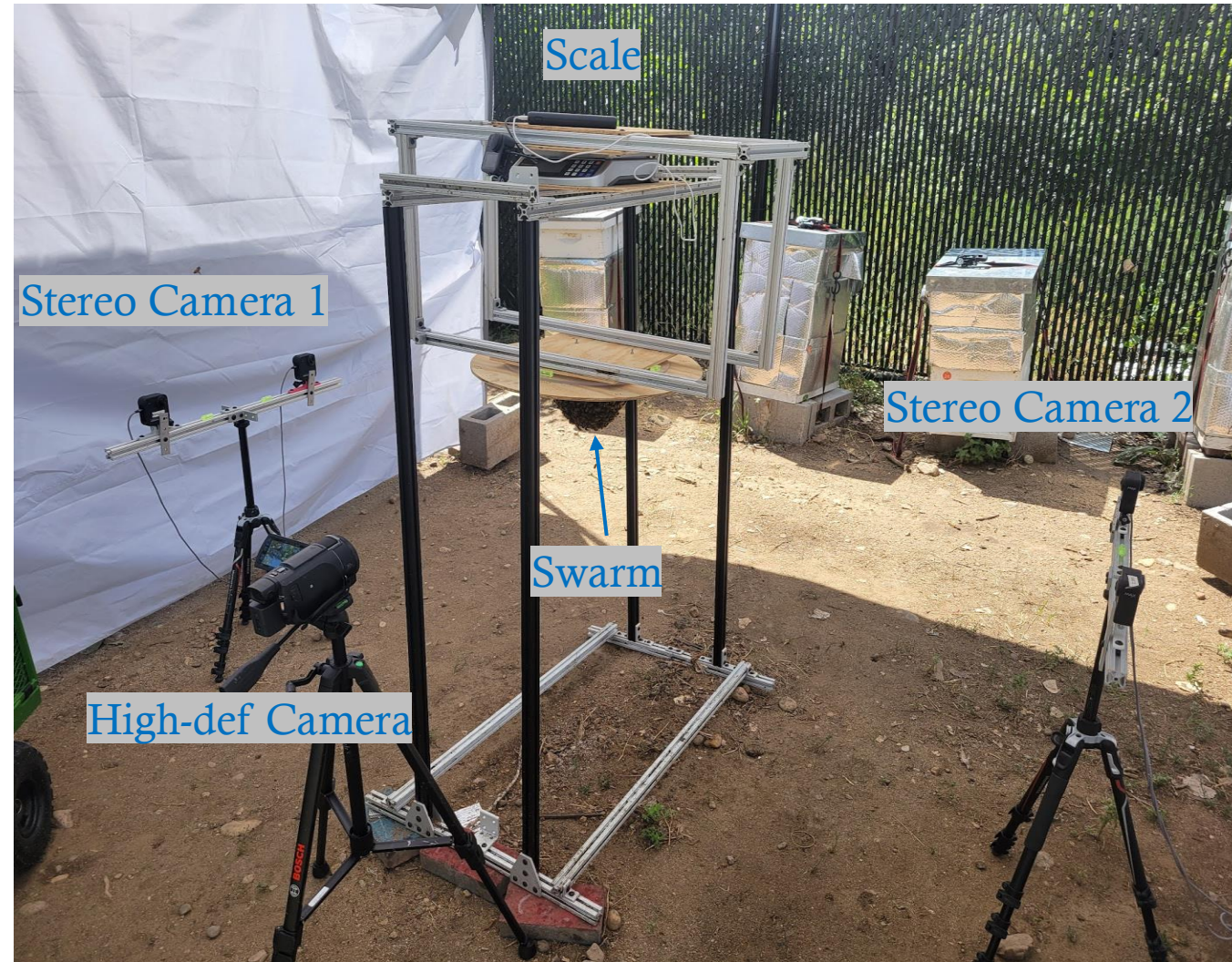
100x Speed Indoor Swarm Example

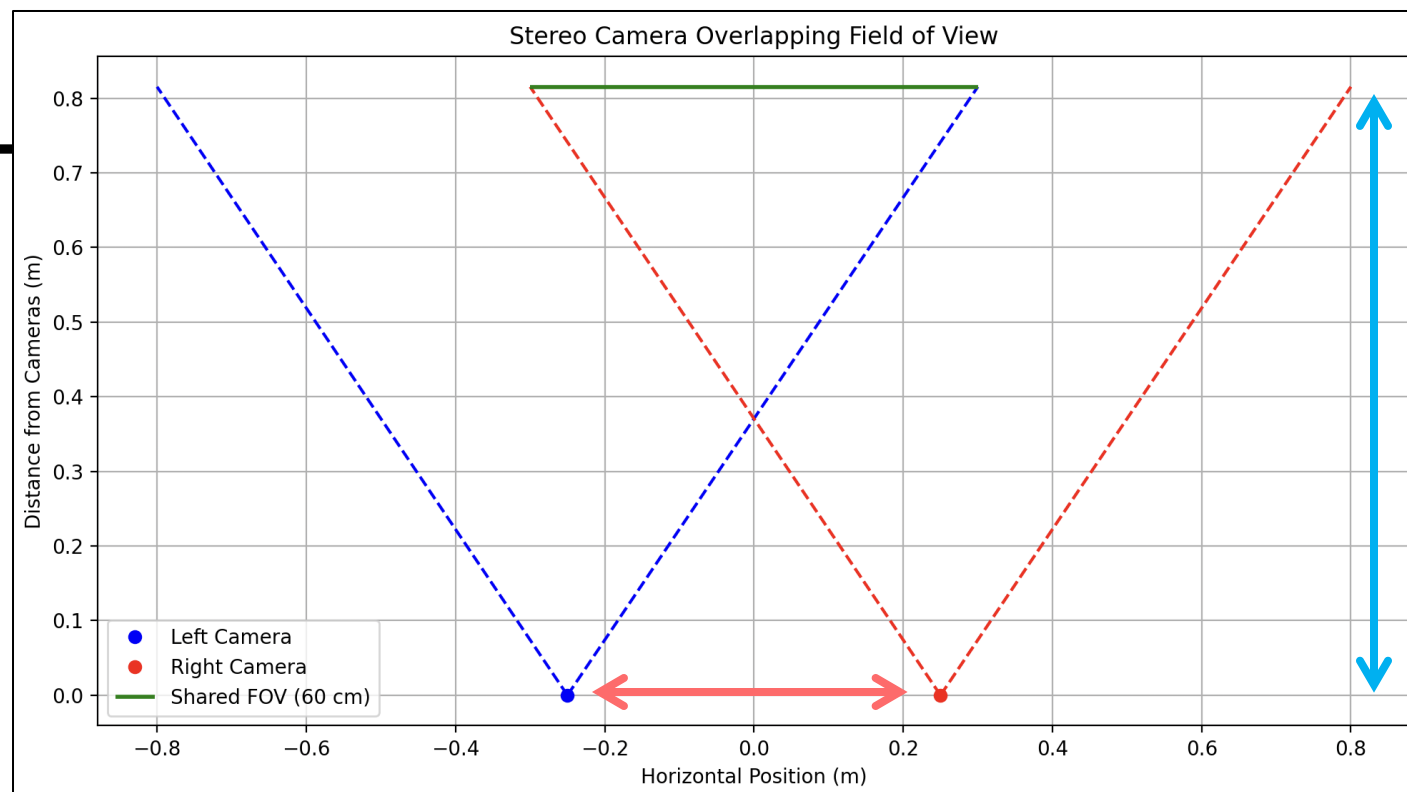
EXPERIMENT DESIGN

How can we design an experiment to capture this?

What we want:

- Protect Swarm
 - Able to leave swarm overnight
 - protect from weather & sun
- Track movement of flying bees in 3d
 - Reconstruct a semi-accurate model of the swarm





	Distance Away (m)			
Distance Apart (m)	0.25	0.5	0.75	1
0.25	0.09	0.42	0.76	1.1
0.5	-0.16	0.17	0.51	0.85
0.75	-0.41	-0.08	0.26	0.6
resolution (pixels/cm)	42.7	21.35	14.23	10.67

Overlapping FOV



“Incentivized” Swarm



Swarm Reformation



40x Speed Outdoor Swarming Example

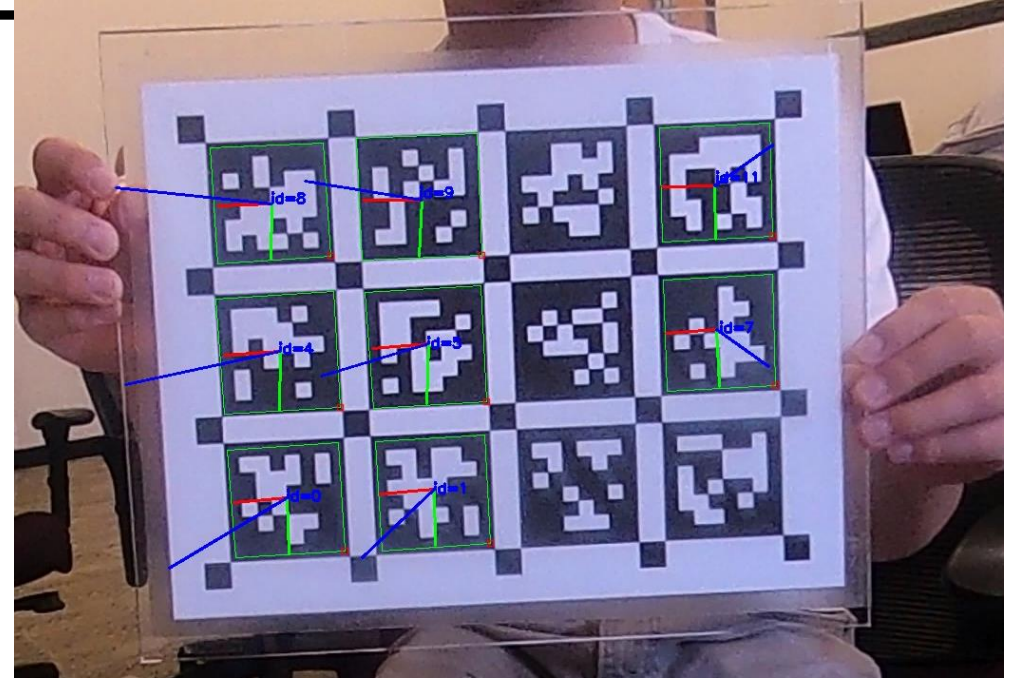
BRIEF OBSERVATIONS (CONCLUSION)

- Swarms do behave differently in the wild.
- At high temperatures (30°C), swarms stay semi-clumped, whereas indoors, bees start crawling away from the swarm
- Flying swarm reconstruction



FUTURE WORK

- Collect more data
 - Incentivize swarming through fake houses
- Process & analyze data
 - Calibrate cameras / undistort images
 - I'm currently working on
 - Stitch left-right view together
 - Reconstruct a 3d model using our sets of cameras and see if the shape matches visually
 - Develop a method to track a bee as it flies through time
 - Being worked on by a grad student right now.



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